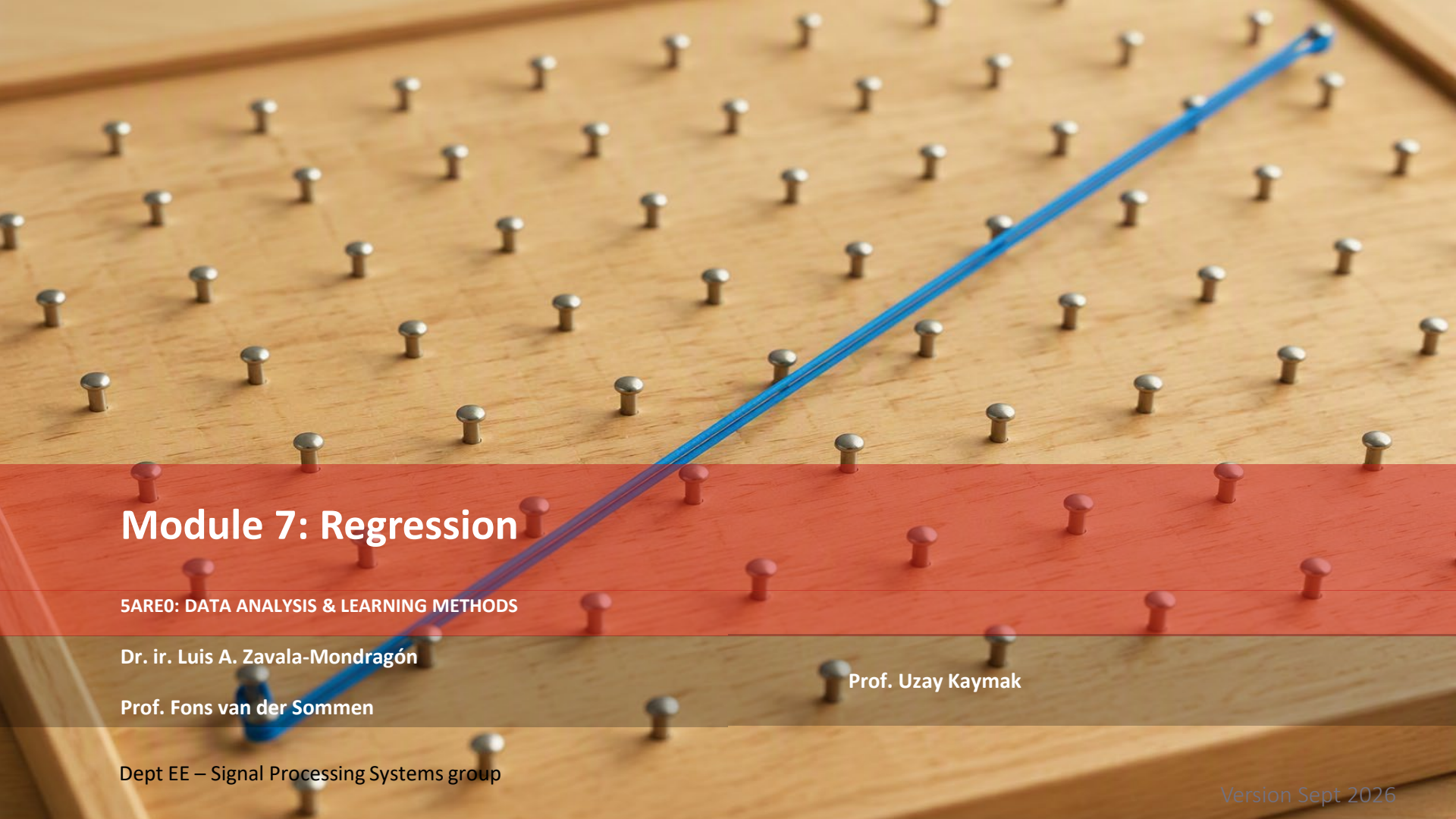




Module 7: Regression

SARE0: DATA ANALYSIS & LEARNING METHODS

Dr. ir. Luis A. Zavala-Mondragón

Prof. Fons van der Sommen

Prof. Uzay Kaymak

Dept EE – Signal Processing Systems group

Version Sept 2026

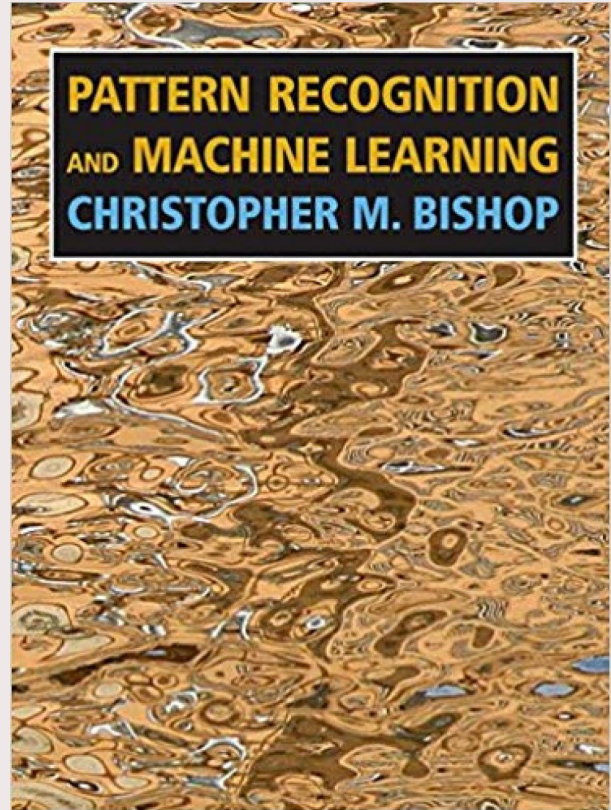
Last time

- KNN classifier
- Logistic regression
- Decision trees

Notation

Established text on ML. Also will be used for other contents.

The content for this module is based on Chapter 3.



Mathematical notation

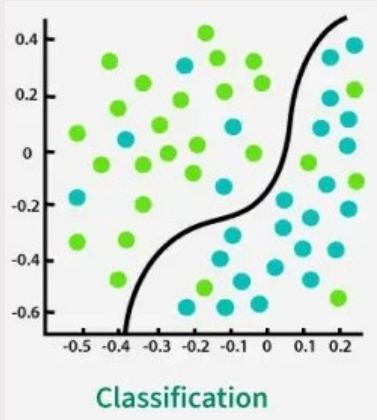
Symbol	Description
$x, \mathbf{x}, \mathbf{X}$	Scalar variable, vector and matrix
(x_i, t_i)	i -th data point
$y(\cdot)$	Regression model
$\mathbf{w}$	Weights of the regression model
$\phi_i(x), \phi_i, \Phi$	Basis function, short notation for i -th basis function, array of bases
$\mathcal{L}(\cdot)$	Loss function

Learning objectives for this lecture

1. Summarize the concept of linear regression and how it differs from classification.
2. Understand what are mappings in linear regression and how they allow for non-linear regression.
3. Apply the techniques for estimating model weights for linear regression.
4. Differentiate Ridge and LASSO regularization and explain how they control model complexity.
5. Explain how logistic regression differs from (non)linear regression models.

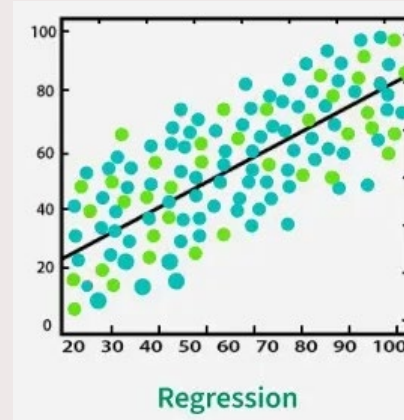
Classification

- Predict discrete numerical output based on the data.
- Model fit finds plane that separates data into discrete categories



Regression

- Predict continuous numerical output based on the data
- Model fit finds plane that is the closest to the data points



Types of regression

- **Linear**
- **Polynomial**
- **Ridge, LASSO, Elastic Net**
- **Non-linear regression and tree-based regression**

Regression

Definition of regression

Under Gaussian additive noise assumption, a regression model can be defined by

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon$$

- In which t is an observed point, $y(\mathbf{x}, \mathbf{w})$ is the data model defined by

$$y(\mathbf{x}, \mathbf{w}) = \sum_{m=0}^{M-1} \phi_m(\mathbf{x}) w_m = \mathbf{\Phi} \mathbf{w}$$

- Vector $\mathbf{x}$ has dims. $N \times 1$ is the independent variables
- Vector $\mathbf{w}$ has dims. $M \times 1$ are the model weights, $w_{m \in \mathbf{w}}$ are m -th model weights
- $\phi_m(x) \in \mathbf{\Phi}$ m -th basis function, $\mathbf{\Phi}$ is called **design matrix**, has dimensions $N \times M$

1M euro question on linear regression

Is the model in the equation

$$y(x, \mathbf{w}) = \sum_{m=0}^{M-1} \phi_m(x) w_m = \mathbf{\Phi} \mathbf{w}$$

Is y linear with respect to x ?

Examples of basis functions $\phi(\cdot)$

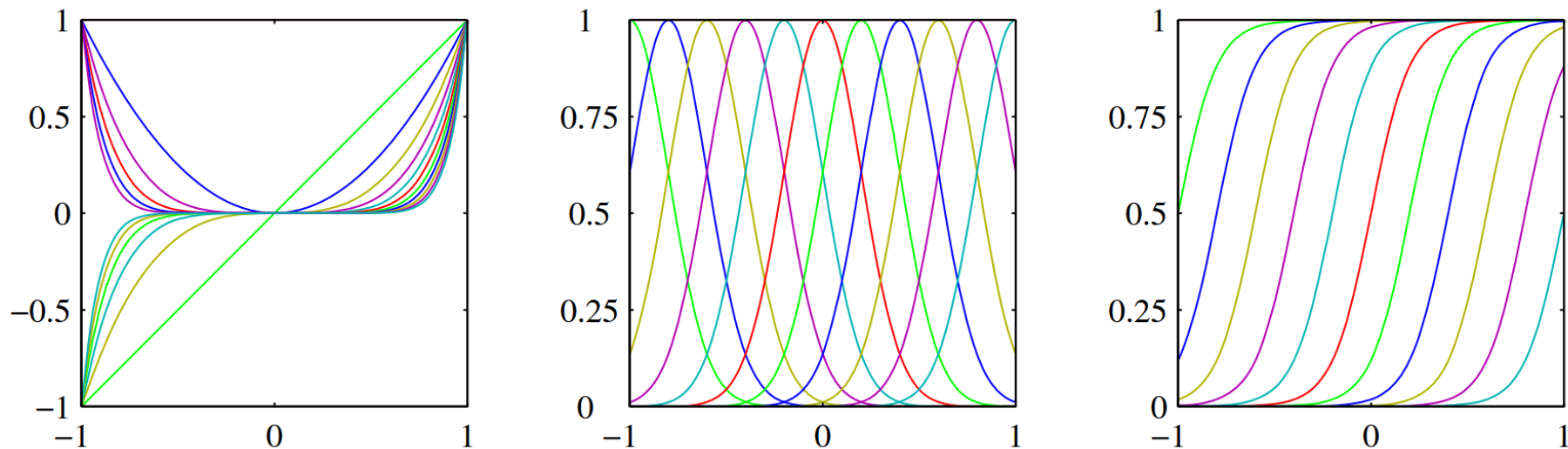


Figure 3.1 Examples of basis functions, showing polynomials on the left, Gaussians of the form (3.4) in the centre, and sigmoidal of the form (3.5) on the right.

Simple example of linear regression

$$t_i = w_0 + w_1 x_i + \epsilon_i$$

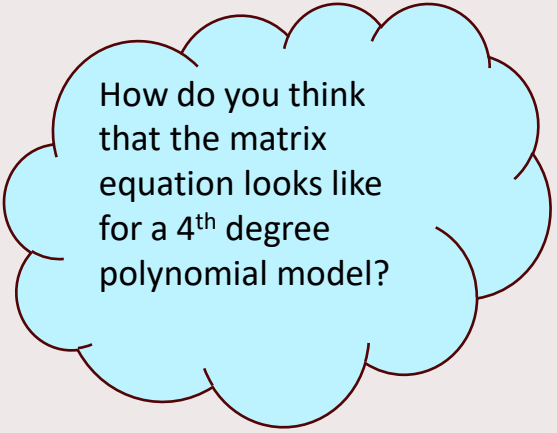
Here, $\phi_m(x) = x^m$.

If we have N_o observations, this becomes

$$\begin{pmatrix} t_0 \\ \vdots \\ t_{N_o-1} \end{pmatrix} = \begin{pmatrix} 1 & x_0 \\ \vdots & \vdots \\ 1 & x_{N_o-1} \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} + \begin{pmatrix} \epsilon_0 \\ \vdots \\ \epsilon_{N_o-1} \end{pmatrix}$$

Or equivalently

$$\mathbf{t} = y(\mathbf{x}, \mathbf{w}) + \epsilon = \mathbf{\Phi} \mathbf{w} + \epsilon$$



How do you think that the matrix equation looks like for a 4th degree polynomial model?

Linear/Polynomial regression

If we allow o model now to have M dimensions

$$t_i = w_0 + w_1 x_i + w_2 x_i^2 + \cdots w_{M-1} x_i^{M-1} + \epsilon_i$$

Where $\phi_m(x) = x^m$. Note in matrix form that this is still formulated as

$$\mathbf{t} = \mathbf{\Phi} \mathbf{w} + \boldsymbol{\epsilon}$$

How can we estimate the model weights $\mathbf{w}$ of model $y(\mathbf{w}, \cdot)$, such that we can predict new unseen values?

Least square estimation (LSE)

To learn the weights of the regression model, we can employ the optimization procedure

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} (\mathbf{t} - \Phi \mathbf{w})^T (\mathbf{t} - \Phi \mathbf{w})$$

This is known as the least-squares estimator.

Note that there is not an explicit modeling of the distribution of the error ϵ .

Polynomial regression with LSE

We solve the linear regressor by minimizing

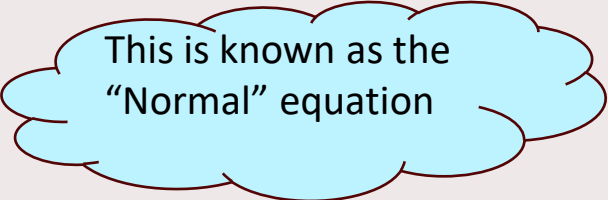
$$\mathcal{L}(\mathbf{w}) = \frac{1}{2} (\mathbf{t} - \Phi \mathbf{w})^T (\mathbf{t} - \Phi \mathbf{w})$$

Expanding this, we have that

$$\mathcal{L}(\mathbf{w}) = \mathbf{w}^T \Phi^T \Phi \mathbf{w} - \frac{1}{2} \mathbf{w}^T \Phi^T \mathbf{t} - \frac{1}{2} \mathbf{t}^T \Phi \mathbf{w} + -\mathbf{t}^T \mathbf{t}$$

Solving $\frac{\partial}{\partial \mathbf{w}} \mathcal{L}(\mathbf{w}) = 0$ with respect to $\mathbf{w}$

$$0 = \Phi^T \Phi \mathbf{w} - \Phi^T \mathbf{t}$$



This is known as the
“Normal” equation

Therefore the estimate of the model weights are

$$\hat{\mathbf{w}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t}$$

Maximum likelihood estimation

The least squares estimator does not model explicitly the distribution of the observations

$$t_n = y(\phi_n, \mathbf{w}) + \epsilon_n$$

Assume that $t_n \sim \mathcal{N}(\mathbf{y}(\phi_n, \mathbf{w}), \sigma_\epsilon)$ and that the samples independent and identically distributed.

We want to estimate the wights of $\mathbf{w}$ that maximize the likelihood that we observe a sample $t \in \mathbf{t}$

Maximum likelihood estimation

Based on the previous slide, the likelihood of observing t is

$$P(t|\mathbf{w}) = k \cdot e^{-\frac{(t-\phi\mathbf{w})^2}{2\sigma_\epsilon^2}}$$

We want to find $\mathbf{w}$ that maximizes the likelihood for N_o observations

$$P(t_0, \dots, t_{N_o-1}|\mathbf{w}) = k^{N_o} \cdot \prod_{i=0}^{N_o-1} e^{-\frac{(t_i-\phi_i\mathbf{w})^2}{2\sigma_\epsilon^2}}$$

Which is equivalent to

$$P(t_0, \dots, t_{N_o-1}|\mathbf{w}) = k^{N_o} \cdot e^{-\frac{1}{2\sigma_\epsilon^2} \sum_{i=0}^{N_o-1} (t_i - \phi_i\mathbf{w})^2}$$

Maximum likelihood estimation

Rather than maximizing the likelihood, we can minimize the neg. log-likelihood

$$-\log \left(P(t_i, \dots, t_{N_o-1} | \mathbf{w}) \right) = \frac{1}{2\sigma_\epsilon} \sum_{i=0}^{N_o-1} (t_i - \boldsymbol{\phi}_i^T \mathbf{w})^2 - N_o \log(k)$$

Expressing everything in vector form

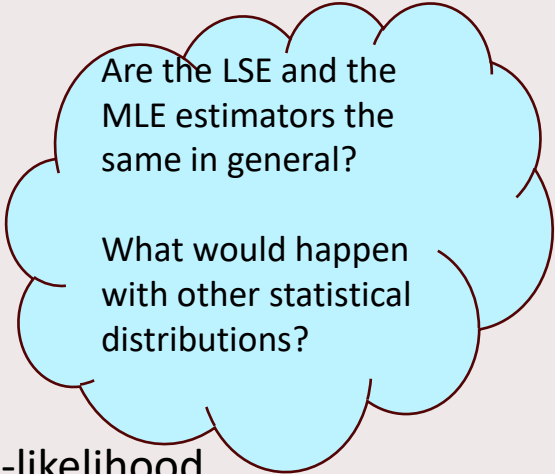
$$J = -\log \left(P(t_i, \dots, t_{N_o-1} | \mathbf{w}) \right) = \frac{1}{2\sigma_\epsilon} \underbrace{(\mathbf{t} - \mathbf{W}^T \boldsymbol{\Phi})^T (\mathbf{t} - \mathbf{W}^T \boldsymbol{\Phi})}_{\text{LSE}} + k$$

Maximum likelihood estimation

$N_o \log(k)$ does not depend on $\mathbf{w}$. Then J minimizing the negative log likelihood is equivalent to minimize the MSE!

Then the estimated $\hat{\mathbf{w}}$ weights are also defined by

$$\hat{\mathbf{w}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t}$$



Are the LSE and the MLE estimators the same in general?

What would happen with other statistical distributions?

Note that the least squares estimator (LSE) and the maximum-likelihood estimator (MLE) are the same because the model assumes that the noise is **Gaussian**

Example 1: Polynomial regression

True signal

$$y = 1 + 2x - 3x^2$$

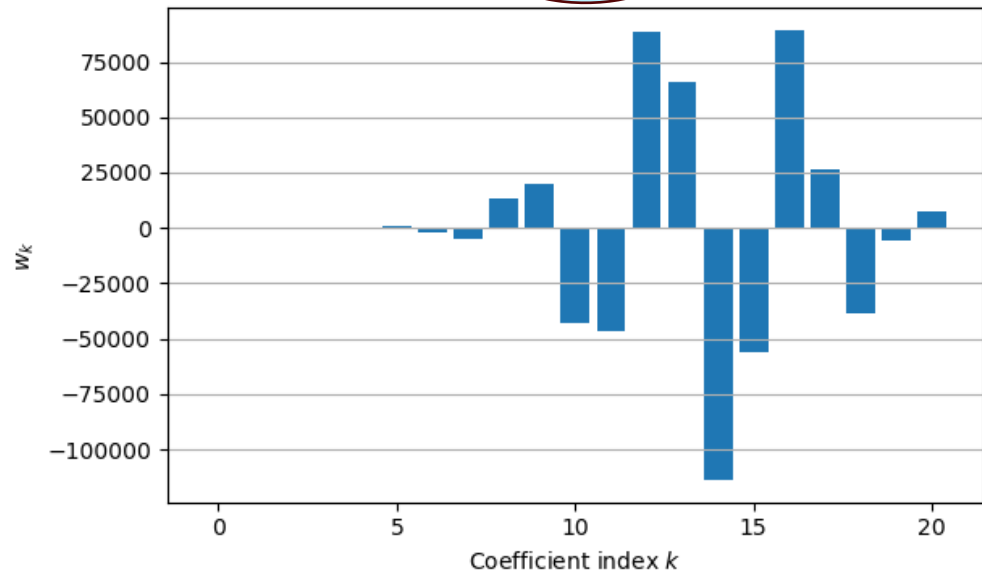
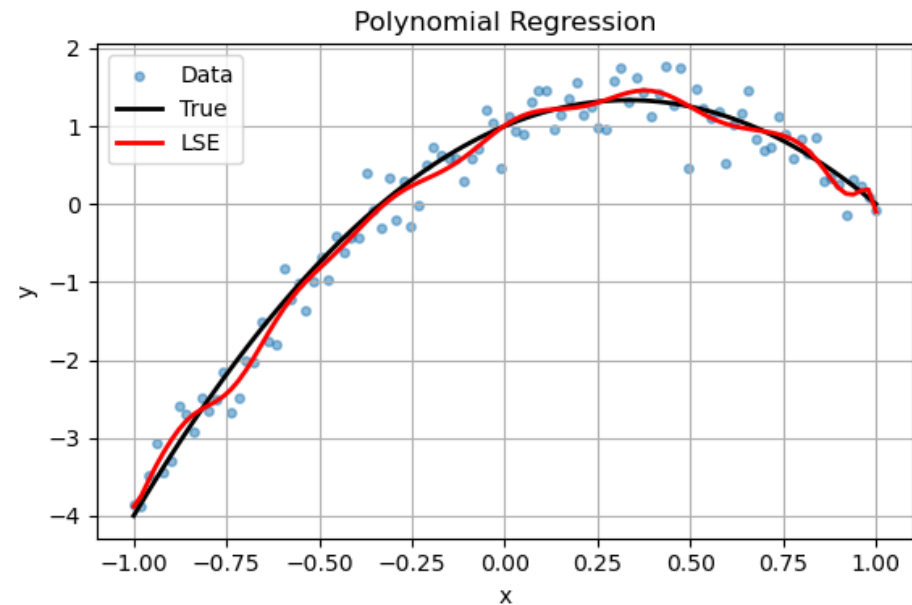
100 observed samples from

$$y_{\text{observed}} = 1 + 2x - 3x^2 + \eta$$

Let's perform a 20th order regression

Example 1: data fitting

What can you say about the estimated coefficients?



$$y_{\text{observed}} = 1 + 2x - 3x^2 + \eta$$

Example 2: Fourier transform

- Assume that we have 250 samples acquired
- The signal contains Two sinusoidal functions. One at 10 Hz and one at 40Hz.
- The signal is also contaminated with noise.
- Can we estimate the signal content with linear regression?

Example 2: Signal model

We assume $\phi_m = a_k \cos\left(\frac{2\pi m}{N} x\right) + j b_k \sin\left(\frac{2\pi m}{N} x\right)$. The “linear” model is defined by

$$y = \Phi\beta + \epsilon,$$

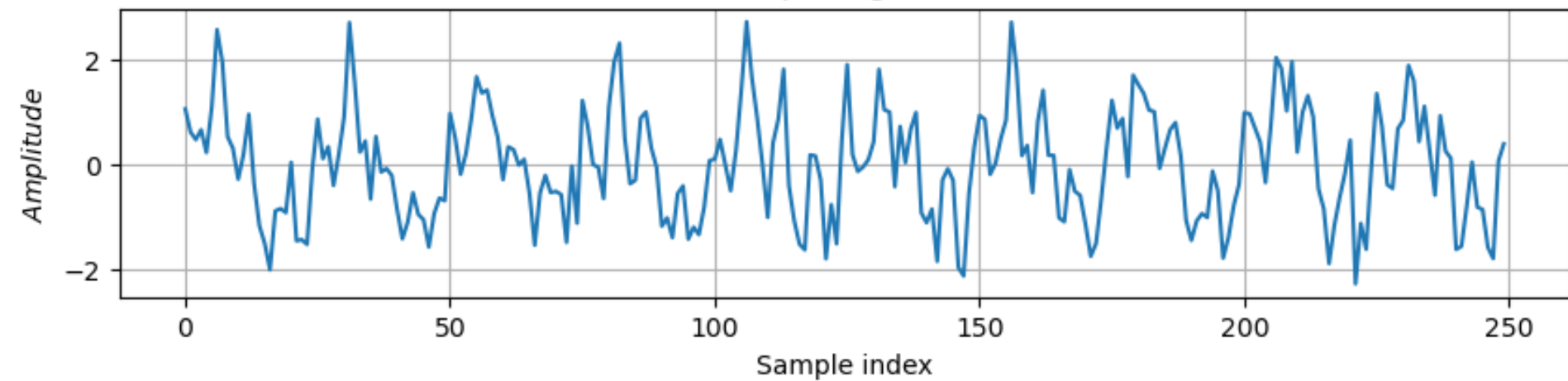
Where

$$\Phi = \begin{pmatrix} 1 & \cos(\omega_1 x_1) & \sin(\omega_1 x_1) & \dots & \cos(\omega_M x_1) & \sin(\omega_M x_1) \\ 1 & \cos(\omega_1 x_2) & \sin(\omega_1 x_2) & \dots & \sin(\omega_M x_2) & \sin(\omega_M x_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & \cos(\omega_1 x_D) & \sin(\omega_1 x_2) & \dots & \sin(\omega_M x_2) & \sin(\omega_M x_D) \end{pmatrix}$$

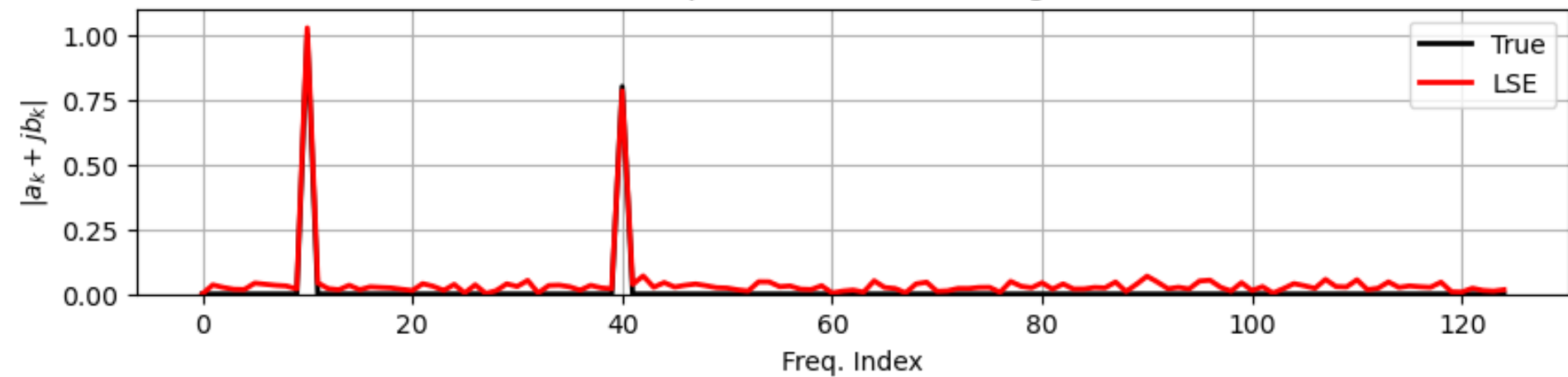
and

$$\beta = (a_0 \quad a_1 \quad b_0 \quad \dots \quad a_M \quad b_M)^T$$

Input signal



Fourier Spectrum via Linear Regression



Numerical solutions

If the dataset is very large, it is expensive to compute the closed-form solution

$$\hat{\mathbf{w}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t}.$$

Alternatively, we can solve the optimization problem numerically with methods such as steepest gradient descent, conjugate least squares (CGLS), etc. The simplest is the SGD solution given by

$$\hat{\mathbf{w}}_{n+1} = \hat{\mathbf{w}}_n - \alpha \Phi^T (\mathbf{t} - \Phi \mathbf{w})$$

Bias-variance tradeoff

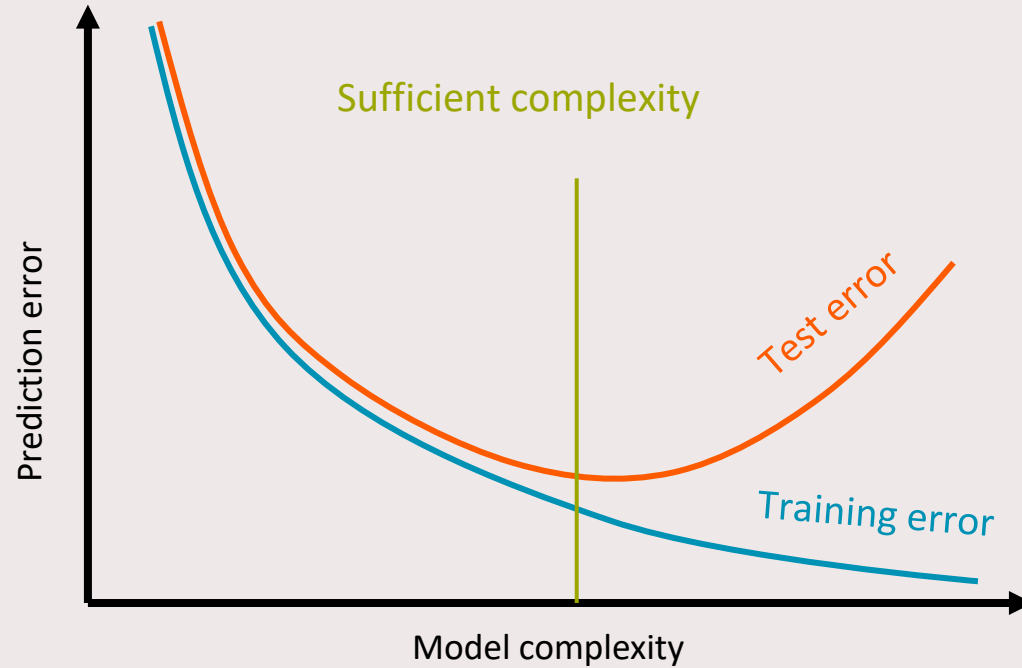
Bias-variance trade-off

The MSE and MLE find weights with the minimum average error for a dataset.

More degrees of freedom = model learns more complex relationships

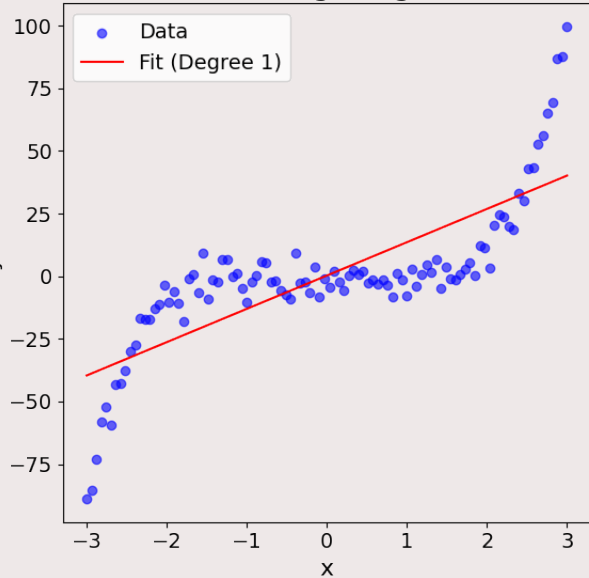
- **Bias error:** *Caused by erroneous assumptions in the model.*
 - *High bias -> Model misses relevant relations between feats. and targs. (underfitting).*
- **Variance error:** *High sensitivity to small fluctuations in the training set.*
 - *High variance -> Model captures random fluctuations in the training data (overfitting)*

Bias-variance trade-off

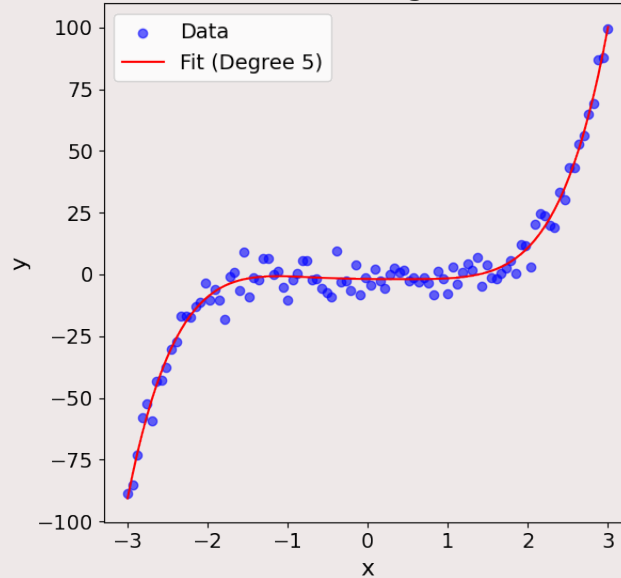


Bias-variance trade-off in practice

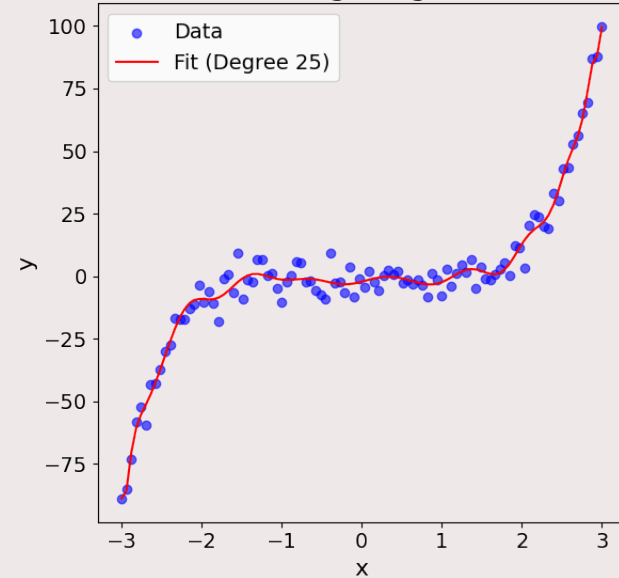
Underfitting (Degree 1)



Good Fit (Degree 5)



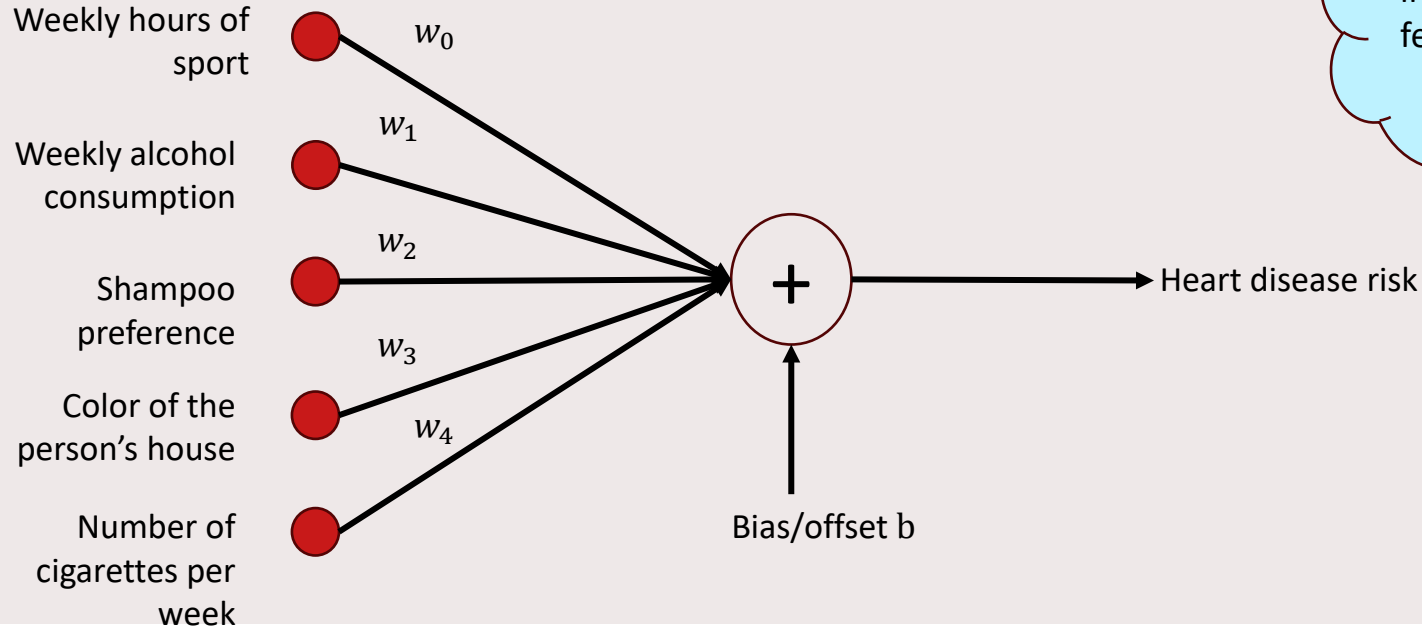
Overfitting (Degree 25)



Regularization

Are all features equally important?

Assume that you have a model that predicts heart disease risk



What would you do to irrelevant features?

The concept of regularization

Higher complexity reduces model bias up to a point. **BUT, the model should not be more complex than needed!**

How can we define how complex the model should be? To avoid overfitting, we can penalize the complexity of the model

$$\mathcal{L} = \text{MSE}(\mathbf{t}, y(\mathbf{x}, \mathbf{w})) + \lambda R(\mathbf{w})$$

Ridge regression

In order to reduce the model complexity, we can reduce the magnitude of the coefficients. Let assume

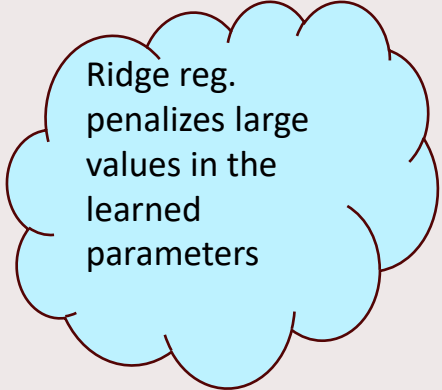
$$R(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

This leads to the loss

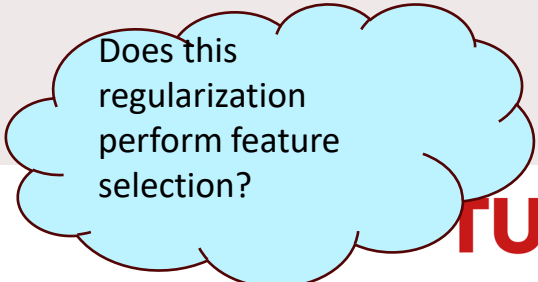
$$\mathcal{L} = \text{MSE}(\mathbf{t}, y(\mathbf{x}, \mathbf{w})) + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

The value of $\mathbf{w}$ that minimizes $\mathcal{L}$ leads to the model weights

$$\hat{\mathbf{w}} = (\lambda \mathbf{I} + \Phi^T \Phi)^T \Phi^T \mathbf{t}$$

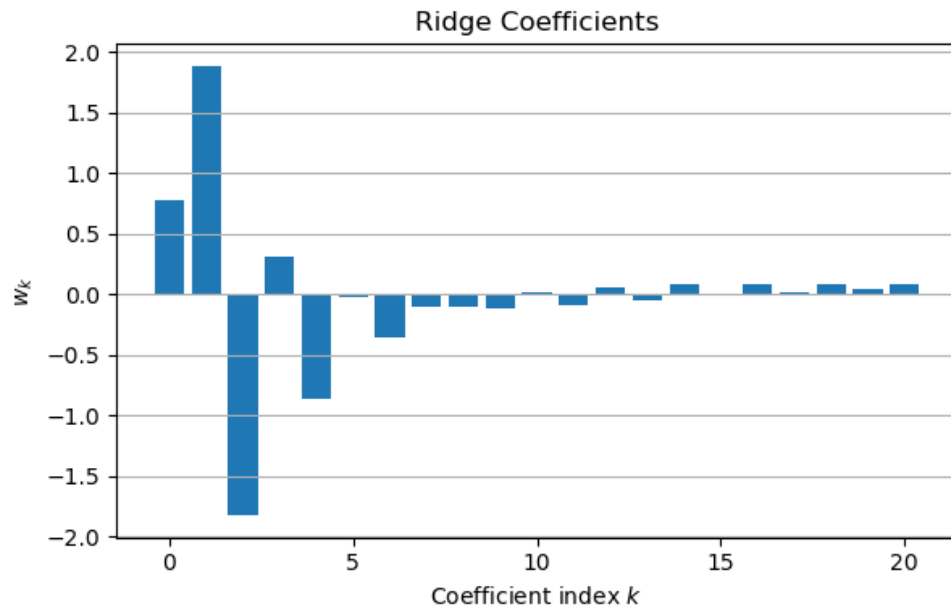
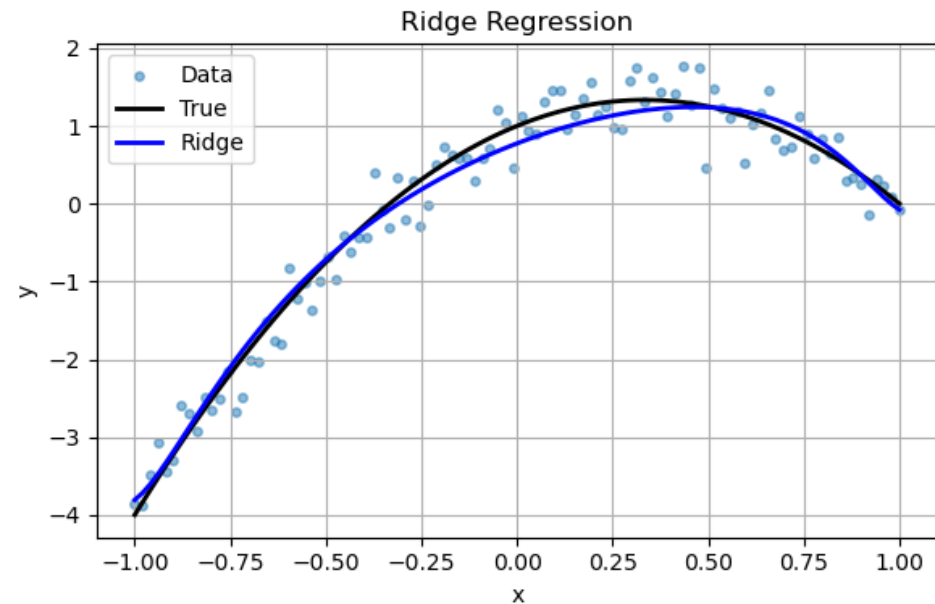


Ridge reg.
penalizes large
values in the
learned
parameters



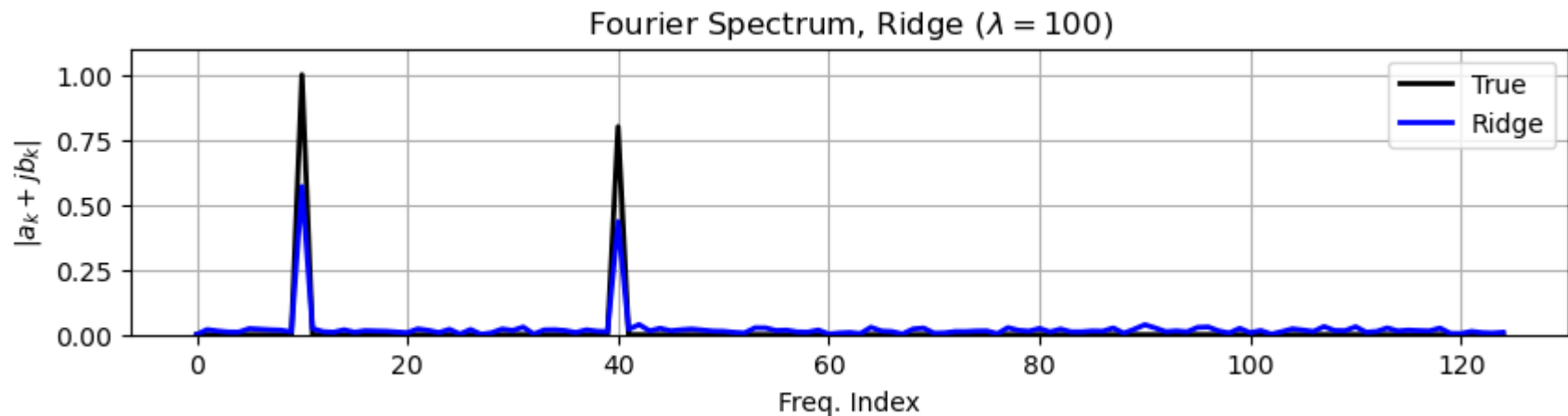
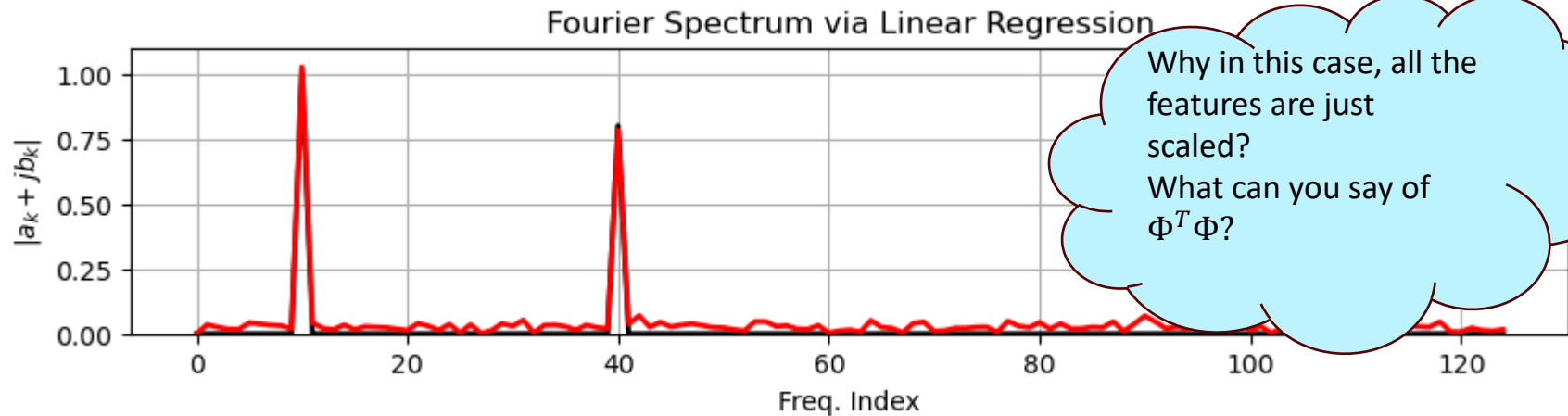
Does this
regularization
perform feature
selection?

Example 1: Ridge regression



$$y_{\text{observed}} = 1 + 2x - 3x^2 + \eta$$

Going back to example 2



LASSO

Rather than the L2 norm, LASSO penalizes the magnitude of coefficients $\mathbf{w}$

$$R(\mathbf{w}) = \|\mathbf{w}\|_1$$

This leads to the loss

$$\mathcal{L} = \text{MSE}(\mathbf{t}, y(\mathbf{X}, \mathbf{w})) + \lambda \|\mathbf{w}\|_1$$

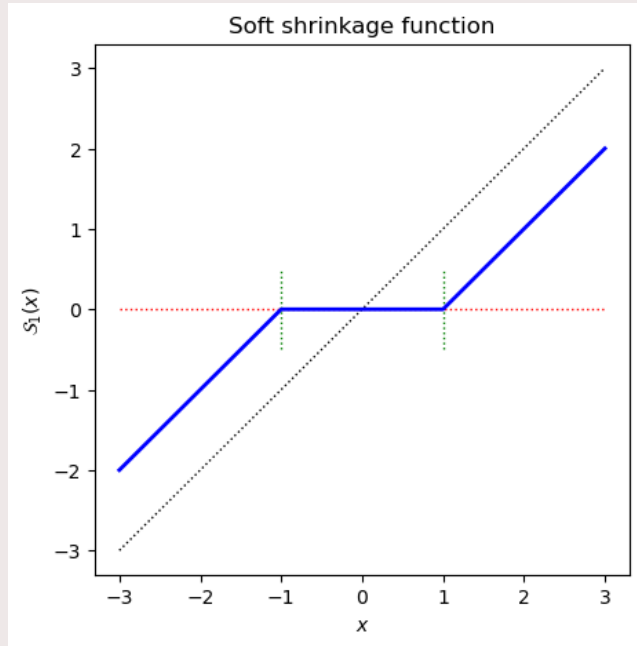
The value of $\mathbf{w}$ that minimizes $\mathcal{L}$ leads to the model weights for ORTHOGONAL DESIGN MATRICES ($\Phi^T \Phi = \mathbf{D}$, where $\mathbf{D}$ is diagonal)

$$\hat{\mathbf{w}} = \mathcal{S}_{\lambda/2} \left((\Phi^T \Phi)^{-1} \Phi^T \mathbf{t} \right)$$

If design matrices are not orthogonal, we need to use numerical solutions.

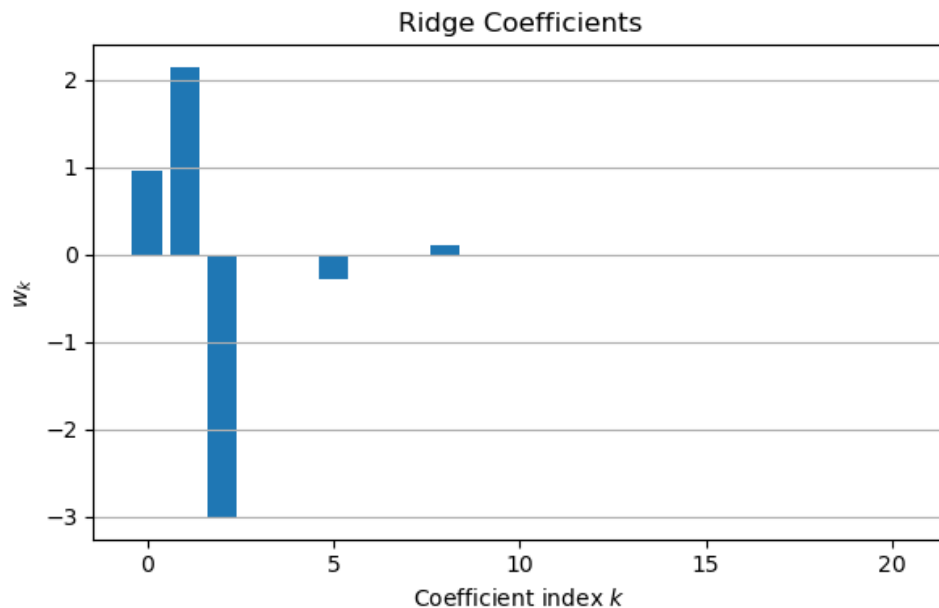
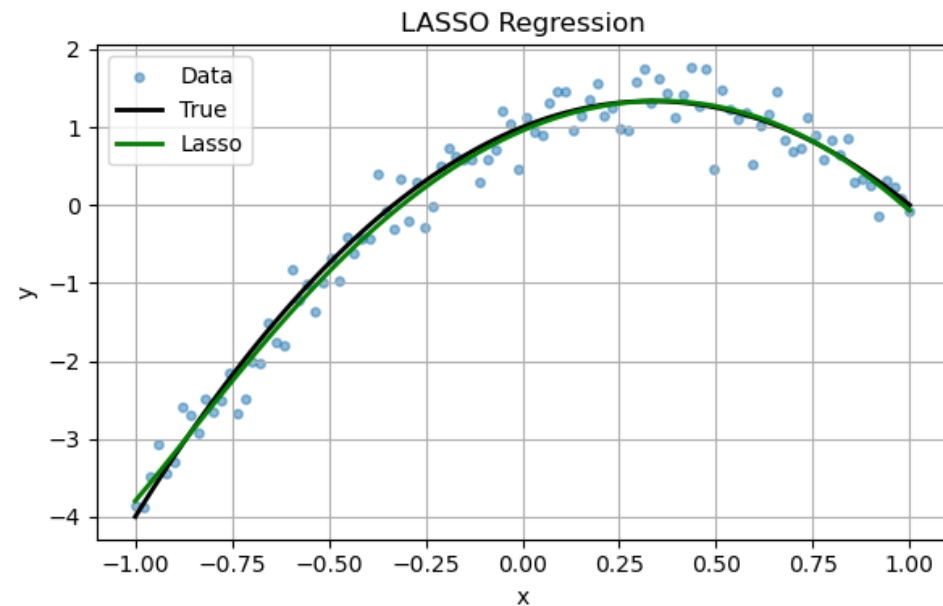
$$\hat{\mathbf{w}}_{n+1} = \mathcal{S}_{\lambda/2} \left(\hat{\mathbf{w}}_n - \alpha \Phi^T (\mathbf{t} - \Phi \mathbf{w}) \right)$$

The soft shrinkage function



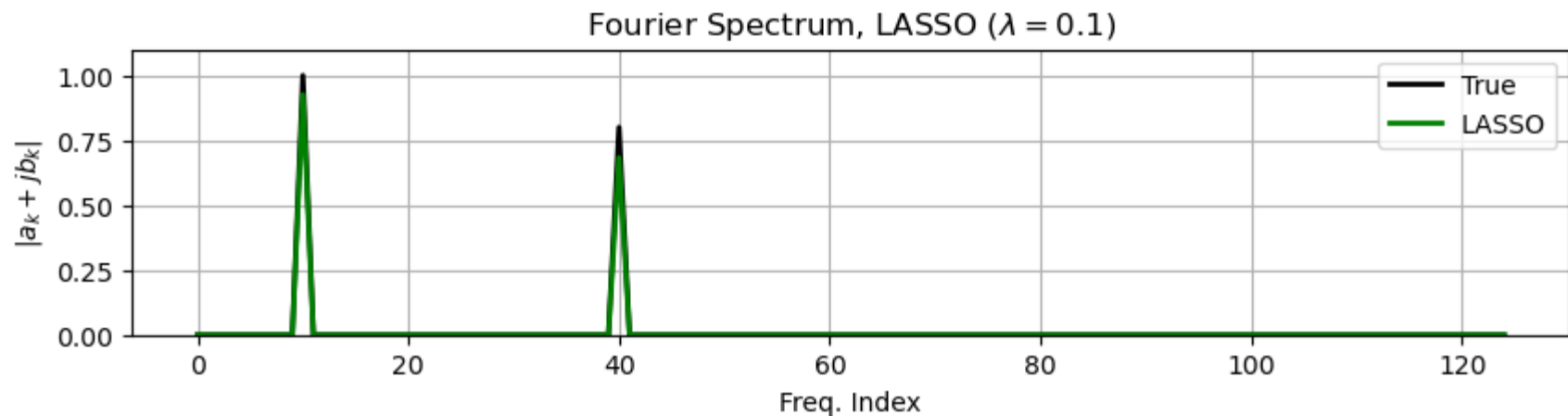
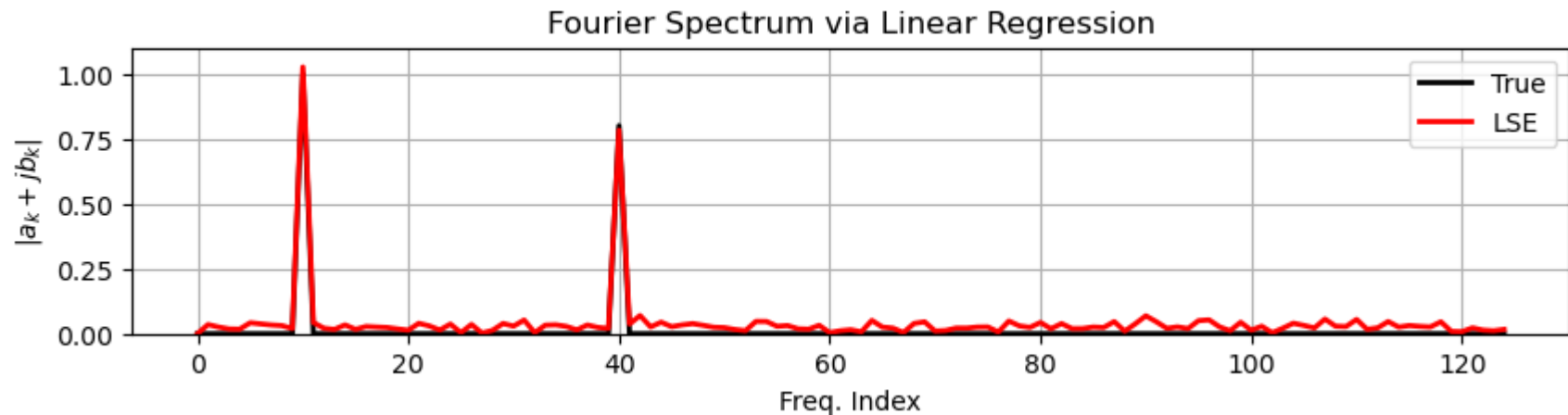
$$S_\lambda(x) = \text{sign}(x) \cdot \max(|x|, -\lambda \ 0)$$

Example 1: LASSO regression

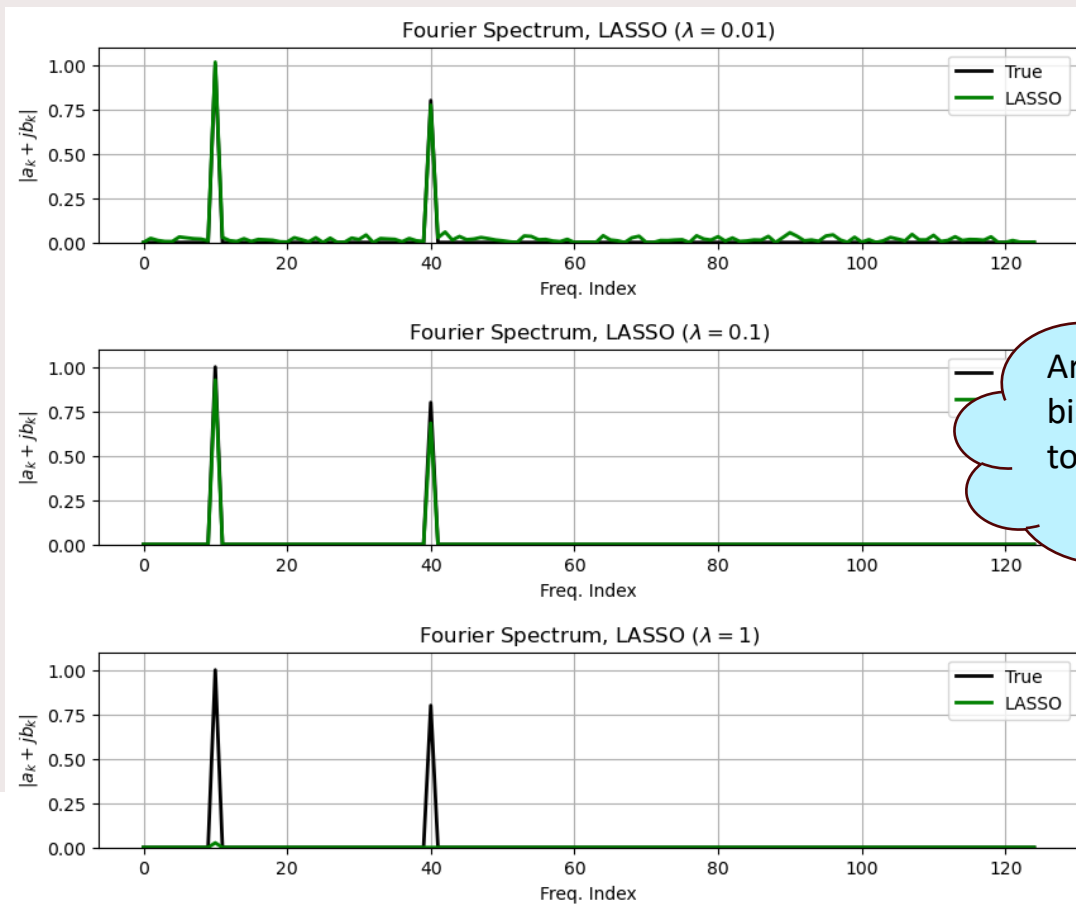


$$y_{\text{observed}} = 1 + 2x - 3x^2 + \eta$$

Example 2: LASSO regression



Bias/variance in regression



Are we introducing bias at the expense to reduce variance?

Recap questions

- **In which conditions would you choose**
 - a) Unregularized regression
 - b) Ridge regression
 - c) LASSO
- **What is the core difference between each of these methods?**

Logistic regression

Logistic regression: Connecting to regression

Logistic regression is actually a **classification** technique. However, its formulation makes it similar to regression models.

The model is defined by

$$p(\mathcal{C}_1|\phi, \mathbf{w}) = y(\phi) = \sigma(\phi^T \mathbf{w})$$

$p(\mathcal{C}_1|\phi)$ is the prob and $\sigma(\cdot)$ is the *sigmoid* function. That input $\phi = (\phi_0(x_0), \dots, \phi_{M-1}(x_{M-1}))$ is of class 1. Alternatively, the class 2 is defined by

$$p(\mathcal{C}_2|\phi, \mathbf{w}) = 1 - p(\mathcal{C}_1|\phi)$$

How is logistic regression trained?

Remember the binary cross entropy

$$\mathcal{L}(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = \sum_{n=0}^{N-1} \{t_n \ln y_n + (1 - t_n) \ln(1 - y_n)\}$$

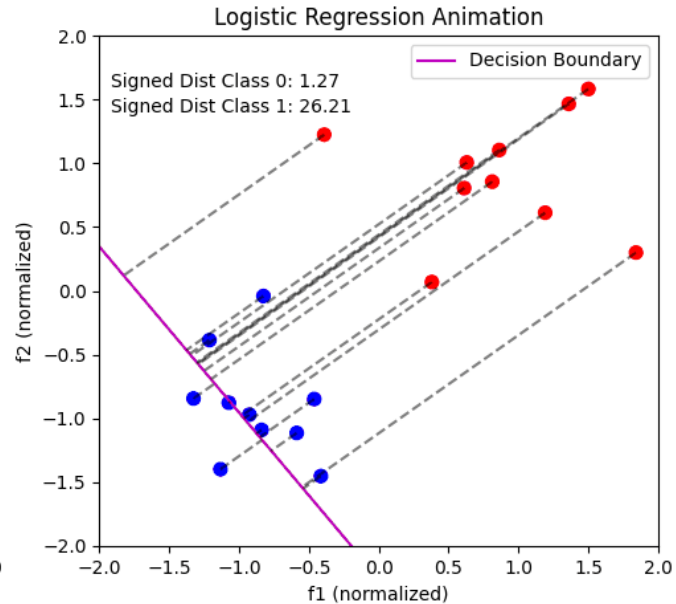
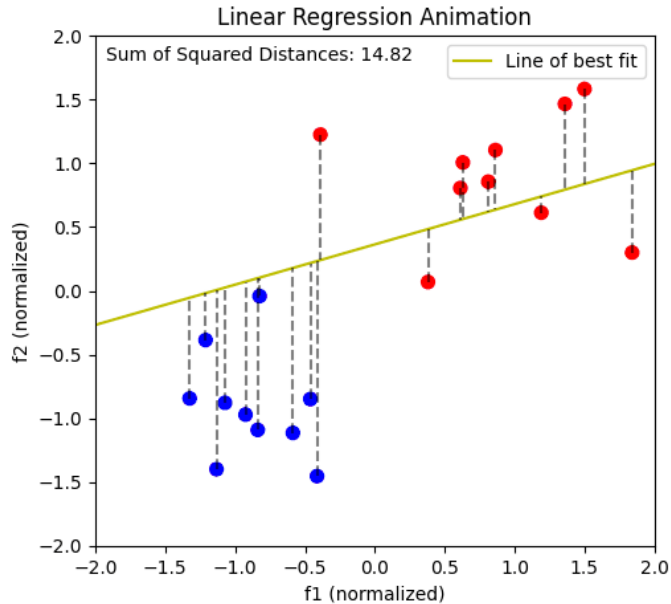
Where $y_n = (\mathcal{C}_n|\phi, \mathbf{w})$. In this case, there is no closed-form solution for an estimate of $\mathbf{w}$. Therefore an iterative algorithm (e.g. SGD) is needed

$$\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha \nabla \mathcal{L}(\mathbf{w}_i)$$

Where the gradient $\nabla \mathcal{L}(\mathbf{w})$ is defined by

$$\nabla \mathcal{L}(\mathbf{w}) = \sum_{n=0}^{N-1} (y_n - t_n) \phi_n$$

Logistic regression vs linear regression



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Summary & look ahead

Summary

- ***What is regression?***
 - Regression models find the (hyper)plane that brings close the estimated function to the observed data.
- **What is the role of a mapping in linear regression?**
 - Mappings allow to map the data in such a way that a non-linear problem can be approximated to a linear one.
- **What is the difference between LSE and MLE?**
 - Least squares estimation: No explicit assumptions about statistical distribution of data
 - MLE: Explicit statistical modeling of data distribution.
 - LSE and MLE equivalent for Gaussian assumption.

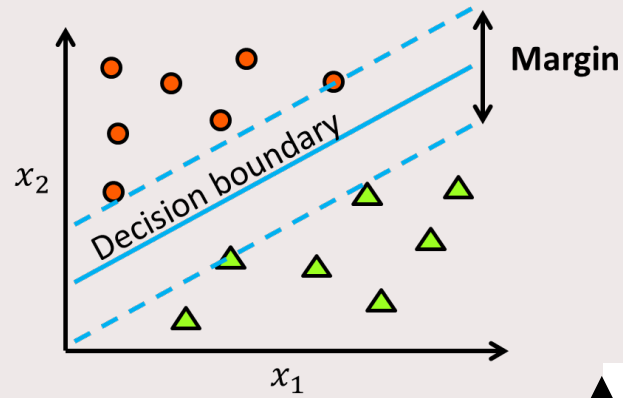
Summary (II)

- ***How do we estimate model weights in regression?***
 - There are different estimators to perform regression. We saw MSE and MLE and SGD.
- ***Bias-variance tradeoff.***
 - High model complexity yields lower training error (low bias), but very complex models have higher errors in the test set (high variance, i.e. *overfitting*). We want to balance bias and variance.
- ***Reducing model complexity.***
 - To reduce model complexity we can use regularization techniques (e.g. Ridge regression, or LASSO).

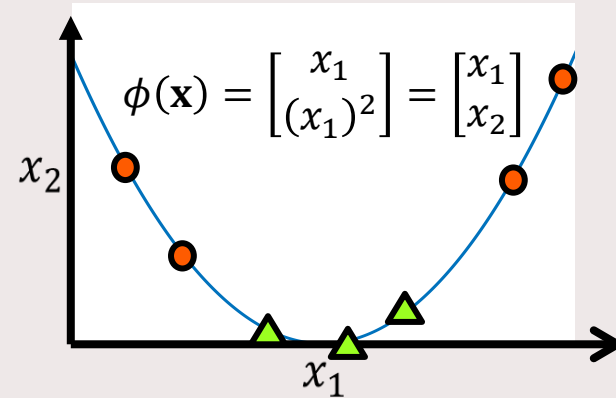
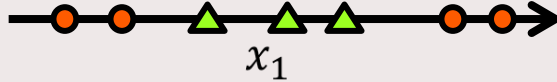
Summary (III)

- ***What is logistic regression?***
 - It is a classification model. It is a linear model where there is a sigmoid non-linearity.
 - Its weight estimator does not have closed form.

Look ahead



Not linearly separable



Questions?

Use of AI

In this presentation, the cover image, as well as the figures in slide 29 were generated with the help of MS 365 copilot professional. After using this tool the validity of the tool's outputs were validated. Luis A. Zavala Mondragón takes full responsibility for the content of the work.